

PROJECT REPORT

Import Data using Transform Maps (Spreadsheet)

A ServiceNow-based solution to import employee training records from a spreadsheet, link them to employees and protect them with role-based access control

Item	Details
Project Name	Import Data using Transform Maps (Spreadsheet)
Team ID	6ab4d8aca238dc999eb5c33f
Platform	ServiceNow
Date	03 October 2026
Project Duration	14 September 2026 to 03 October 2026 (3 Sprints)
Document Type	Final Project Report

TABLE OF CONTENTS

1. INTRODUCTION

1.1 Project Overview

Organisations that use ServiceNow often keep employee training information in spreadsheets maintained by HR teams. Each spreadsheet holds details such as the name of the training, the date on which it was completed, its current status and the employee who attended it. Because these sheets live outside the platform, the data is not connected to employee records, cannot be reported on together with department information and is not protected by the platform's security model.

This project, Import Data using Transform Maps (Spreadsheet), solves that problem inside ServiceNow. A custom table called Employee Training Records is created with four fields: Training Name, Completion Date, Status and Employee. A spreadsheet containing the same four columns is uploaded through the Load Data feature into an import set staging table, and a Transform Map moves the staged rows into the target table, matching each source column to the correct target field.

After the data is imported, the Employee field is dot-walked to show the employee's Department on the form, which supports reporting. Finally, a new HR Manager role is created and Read and Write Access Control Lists (ACLs) are added to the table so that only authorised users can see or change the records. The access rules are verified by impersonating different users.

Aspect	Summary
Problem	Training records stored in spreadsheets, unlinked to employees and unprotected
Solution	Import through Load Data and Transform Map into a custom ServiceNow table
Key Fields	Training Name, Completion Date, Status (In Progress / Completed), Employee
Security	HR Manager role with Read and Write ACLs, elevated with security_admin
Verification	User impersonation to confirm authorised and unauthorised access
Delivery	3 sprints, 12 user stories, 33 story points

1.2 Purpose

The purpose of this project is to give HR managers and administrators a fast, accurate and secure way to bring spreadsheet-based training records into ServiceNow. The specific goals are listed below.

- **Speed:** Replace manual typing of many records with a single bulk upload.
- **Accuracy:** Use a Transform Map so that every spreadsheet column is mapped to the right target field and the data matches the table structure.
- **Traceability:** Link each training record to an employee (sys_user) and expose the Department through dot-walking.

- **Security:** Restrict the table to users holding the HR Manager role using Read and Write ACLs.
- **Assurance:** Prove the access rules work by impersonating authorised and unauthorised users.

Approach	Manual Entry (Before)	Transform Map Import (After)
Time required	High - one record at a time	Low - all rows in one upload
Error risk	High - typing mistakes	Low - field mapping is defined once
Employee link	None	Reference to sys_user
Department reporting	Not available	Available through dot-walking
Access control	None	HR Manager role with ACLs

2. IDEATION PHASE

2.1 Problem Statement

HR managers and administrators receive employee training details in spreadsheets. To use that data in ServiceNow they must either type every record by hand or keep separate Excel files, both of which are slow and error-prone. The records are also not related to employee data and anyone who obtains the file can see the information.

Problem statement: How might we load many training records into ServiceNow quickly and accurately, connect each record to the right employee and department, and make sure that only authorised users can view or edit them?

Question	Answer
I am	An HR manager or ServiceNow administrator
I am trying to	Load employee training records into ServiceNow and report on them by department
But	The data is in spreadsheets, must be typed manually and is not linked to employee records
Because	There is no import mapping, no employee reference and no role-based protection
Which makes me feel	Frustrated, unsure about data errors and worried about data exposure

2.2 Empathy Map Canvas

The empathy map below captures how the primary user, an HR manager, experiences the problem.

Quadrant	Observations
Says	"I have a hundred training records in Excel. Typing them one by one will take hours." "I need to know which department each person belongs to." "Only HR should see this data."
Thinks	Will the dates and statuses match after I enter them? Who else might be able to open this file or table? There must be a faster way to bring the sheet in.
Does	Collects training completion details from employees Maintains separate Excel sheets Discusses data and access needs with the HR team
Feels	Frustrated by repetitive data entry Anxious about mistakes and data exposure
Pains	Time-consuming manual entry Typing errors and mismatched fields No link to employee or department No access control

Quadrant	Observations
Gains	Bulk import in minutes Reliable, mapped data Department-wise reporting Confidence that data is secure

2.3 Brainstorming

The team listed possible ways to solve the problem and then compared them against effort, accuracy and security. The Transform Map import was chosen because it is a standard ServiceNow capability that needs no coding and satisfies all three needs.

Idea	Description	Effort	Accuracy	Security	Decision
Manual entry	Type each record into a form	High	Low	Depends on ACLs	Rejected
Keep Excel files	Continue using separate sheets	Medium	Low	None	Rejected
Load Data + Transform Map	Upload sheet, map fields, transform to target table	Low	High	Add role and ACLs	Selected
Custom script import	Write a script to read the file	High	Medium	Add role and ACLs	Not needed

Group discussion also produced the following supporting ideas that were adopted into the final solution:

- Use a Choice field for Status so that only In Progress and Completed can be stored.
- Use a Reference field for Employee so that each record is tied to a user record.
- Dot-walk to the Department field on the form instead of duplicating department data.
- Create a dedicated HR Manager role and an application menu and module for it.
- Test with impersonation rather than creating many separate test accounts.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey map follows an HR manager from receiving a training spreadsheet to viewing secure, reportable records in ServiceNow.

Stage	Customer Action	Touchpoint	Emotion	Opportunity
Awareness	Receives a new training spreadsheet	Email, Excel	Concerned	Consolidate data in one system
Preparation	Arranges four columns to match the table	Excel / Google Sheets	Careful	Use a clear column template
Import	Uploads file through Load Data	ServiceNow Load Data	Hopeful	Single bulk upload
Mapping	Maps columns with a Transform Map and runs Transform	Transform Maps	Focused	Reusable field mapping
Review	Checks imported records and department on the form	Table list and form	Satisfied	Dot-walked Department
Protection	Assigns HR Manager role and verifies ACLs	Roles, ACLs, impersonation	Confident	Role-based security

3.2 Solution Requirement

Functional requirements describe what the system must do. Non-functional requirements describe how well it must do it.

FR No.	Functional Requirement	Sub-requirement
FR-1	Table creation	Create Employee Training Records with Training Name, Completion Date, Status and Employee
FR-2	Status choices	Provide In Progress and Completed as choices
FR-3	Data upload	Upload the spreadsheet through Load Data into an import set table
FR-4	Field mapping	Create a Transform Map mapping four source fields to target fields
FR-5	Transformation	Run Transform to load staged rows into the target table
FR-6	Employee linking	Dot-walk from Employee to Department on the form
FR-7	Role management	Create the HR Manager role, a backup user and an application menu and module
FR-8	Access control	Create Read and Write ACLs after elevating to security_admin
FR-9	Verification	Impersonate users to confirm allowed and denied access

NFR No.	Non-functional Requirement	Description
NFR-1	Usability	Standard ServiceNow interface; no coding required for HR users
NFR-2	Security	Only HR Manager role can read or write the table
NFR-3	Reliability	Imported values must match the target field types and choices
NFR-4	Performance	A full spreadsheet is processed in a single Transform run
NFR-5	Availability	Data is held on the ServiceNow instance and available when the instance is up
NFR-6	Scalability	The same Transform Map can be reused for later spreadsheets

3.3 Data Flow Diagram

The diagram shows how data moves from the administrator's spreadsheet to the final table and how access is controlled when users request the data.

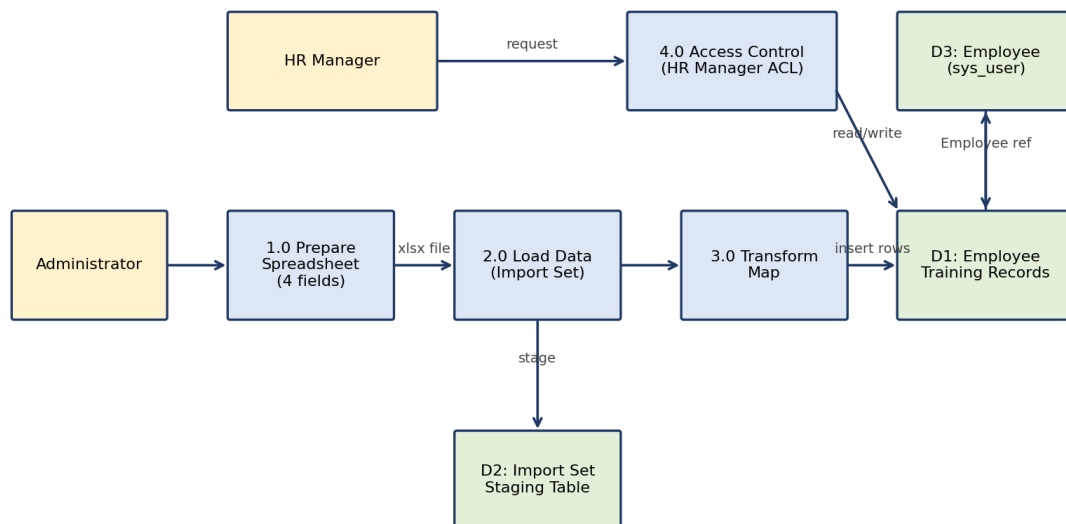


Figure 1: Data Flow Diagram

Process / Store	Description
1.0 Prepare Spreadsheet	Administrator arranges Training Name, Completion Date, Status and Employee columns
2.0 Load Data	File is uploaded and rows are placed in the import set staging table
3.0 Transform Map	Source columns are mapped to target fields and rows are inserted
4.0 Access Control	ACLs check the HR Manager role before read or write is allowed
D1 Employee Training Records	Final table holding the imported records
D2 Import Set Staging Table	Temporary table holding uploaded rows
D3 Employee (sys_user)	Provides the employee reference and department

3.4 Technology Stack

Layer	Technology	Purpose
Platform	ServiceNow instance	Hosts tables, imports, roles and ACLs
Data preparation	Microsoft Excel / Google Sheets	Prepare the four-column spreadsheet
Data import	Load Data and Import Set tables	Stage uploaded rows
Mapping	Transform Maps	Map source fields to target fields
Data model	Custom table with Choice and Reference fields	Store training records
Presentation	Form Layout, application menu and module	Show records and dot-walked Department
Security	Roles, ACLs, security_admin elevation	Restrict access
Testing	User impersonation	Verify access rules

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem-Solution Fit confirms that the solution actually solves the customer's problem. The canvas below summarises the fit for this project.

Canvas Block	Content
1. Customer Segment(s)	HR managers and administrators of organisations that use ServiceNow and keep employee training records in spreadsheets
2. Jobs-to-be-Done / Problems	Load training records quickly into ServiceNow; link each record to an employee and show department for reporting; allow only authorised users to view and edit records
3. Triggers	Training data needs to be consolidated for HR reports or review; a new training spreadsheet is received; need to protect employee information
4. Emotions: Before / After	Before: frustrated, unsure about data errors and data exposure. After: confident, in control and secure
5. Available Solutions	Typing each record manually in ServiceNow; maintaining separate Excel sheets. Pros: simple. Cons: error-prone, no link to employee or department, no access control
6. Customer Constraints	Manual entry of many records is time-consuming; employee data must be visible only to authorised users; spreadsheet data must match the table fields
7. Behaviour	Prepare a spreadsheet with four fields, upload it through Load Data, map fields with a Transform Map, check the records and test access by impersonation
8.1 Channels: Online	ServiceNow web UI (Load Data, Transform Maps, Form Layout, Roles and ACLs); Google Sheets / Excel to prepare the data
8.2 Channels: Offline	Discussing data and access needs with the HR team; collecting training completion details from employees
9. Problem Root Cause	Training data is stored in spreadsheets outside ServiceNow, not related to employee (sys_user) records, and has no role-based protection
10. Your Solution	Import the spreadsheet through Load Data and a Transform Map into the Employee Training Records table; link each record to the employee and dot-walk Department; protect the table with an HR Manager role and Read / Write ACLs; verify by impersonation

4.2 Proposed Solution

The proposed solution combines three ServiceNow capabilities: data import, data relationships and security. Each addresses one part of the root cause.

Root Cause	Solution Component	Outcome
Data outside ServiceNow	Load Data + Transform Map into Employee Training Records	All records in one governed table
No relation to employees	Employee reference field with Department dot-walk	Records tied to people and departments

Root Cause	Solution Component	Outcome
No role-based protection	HR Manager role with Read and Write ACLs	Only authorised users can access data
Uncertainty about correctness	Impersonation testing	Access rules proven before release

Key features of the solution:

- Four-field table design that mirrors the spreadsheet columns exactly.
- Status stored as a choice list, preventing inconsistent text values.
- Reusable Transform Map for future spreadsheet uploads.
- Dedicated application menu and module visible only to the HR Manager role.
- Backup user holding the HR Manager role for continuity.

4.3 Solution Architecture

The architecture is arranged in five layers, from the user interface down to the security rules that guard the data.

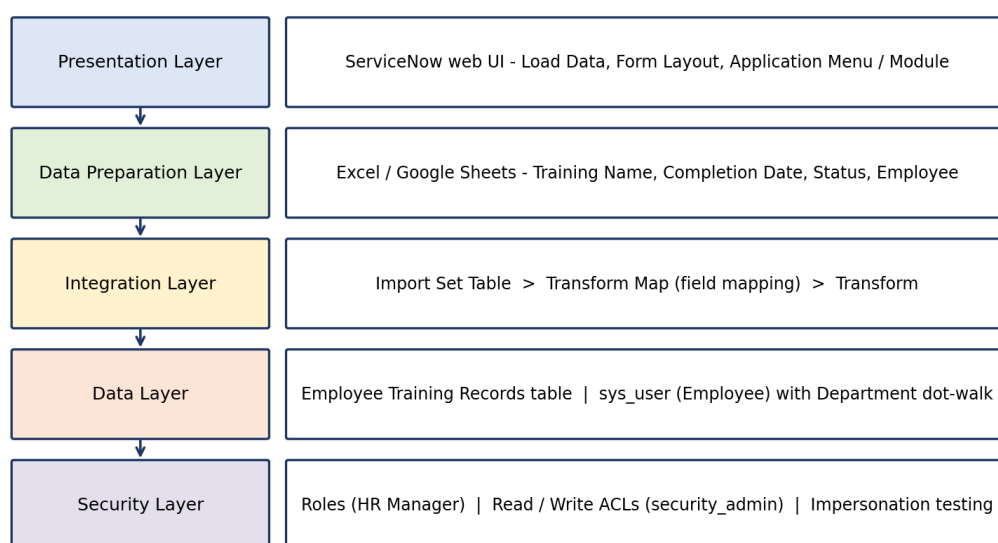


Figure 2: Solution Architecture

Layer	Responsibility
Presentation	Shows the Load Data page, table list, form with dot-walked Department, and application menu
Data Preparation	Excel or Google Sheets file with the four required columns
Integration	Import set staging table and Transform Map that converts staged rows into target records
Data	Employee Training Records table and the Employee (sys_user) reference
Security	HR Manager role, Read and Write ACLs and impersonation checks

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work was organised using Scrum. The product backlog contains twelve user stories grouped into epics and delivered across three sprints of six days each. Story points express relative effort.

Sprint	Epic	USN	User Story / Task	Pts	Priority
Sprint-1	Table Creation	USN-1	As an administrator, I can create the Employee Training Records table with Training Name, Completion Date, Status and Employee fields.	3	High
Sprint-1	Table Creation	USN-2	As an administrator, I can add choices (In Progress, Completed) to the Status field.	1	Medium
Sprint-1	Data Import	USN-3	As an administrator, I can upload a spreadsheet through Load Data into an import set table.	2	High
Sprint-1	Field Mapping	USN-4	As an administrator, I can create a Transform Map and map the four source fields to the target table fields.	3	High
Sprint-1	Field Mapping	USN-5	As an administrator, I can run Transform to load the staged data into the target table.	3	High
Sprint-2	Employee Linking	USN-6	As an administrator, I can dot-walk from Employee to Department on the form.	2	Medium
Sprint-2	Roles & Users	USN-7	As an administrator, I can create a backup user and the HR Manager role and assign the role to the user.	3	High
Sprint-2	Roles & Users	USN-8	As an administrator, I can add the HR Manager role to the application menu and module.	2	Medium
Sprint-2	Access Control	USN-9	As an administrator, I can elevate to security_admin and create Read and Write ACLs for the table with the HR Manager role.	5	High
Sprint-3	Access Control	USN-10	As an HR Manager, I can view and edit Employee Training Records.	3	High
Sprint-3	Access Control	USN-11	As a user without the HR Manager role, I cannot see the Employee Training Records table.	3	High
Sprint-3	Testing	USN-12	As an administrator, I can impersonate users to verify the access rules.	3	Medium

Project tracker:

Sprint	Total Points	Duration	Start Date	Planned End	Points Completed	Actual Release
Sprint-1	12	6 Days	14 Sep 2026	19 Sep 2026	12	19 Sep 2026
Sprint-2	12	6 Days	21 Sep 2026	26 Sep 2026	12	26 Sep 2026
Sprint-3	9	6 Days	28 Sep 2026	03 Oct 2026	9	03 Oct 2026
Total	33	18 Days			33	

Velocity: $\text{Velocity} = \text{Total Story Points Completed} / \text{Number of Sprints} = (12 + 12 + 9) / 3 = 11$ story points per sprint. Average velocity per day $AV = 11 / 6$, which is approximately 1.83 story points per day.

Burndown chart: A burndown chart shows work left to do against time. The actual line stays close to the ideal line, which shows that the team delivered every sprint on its planned date.

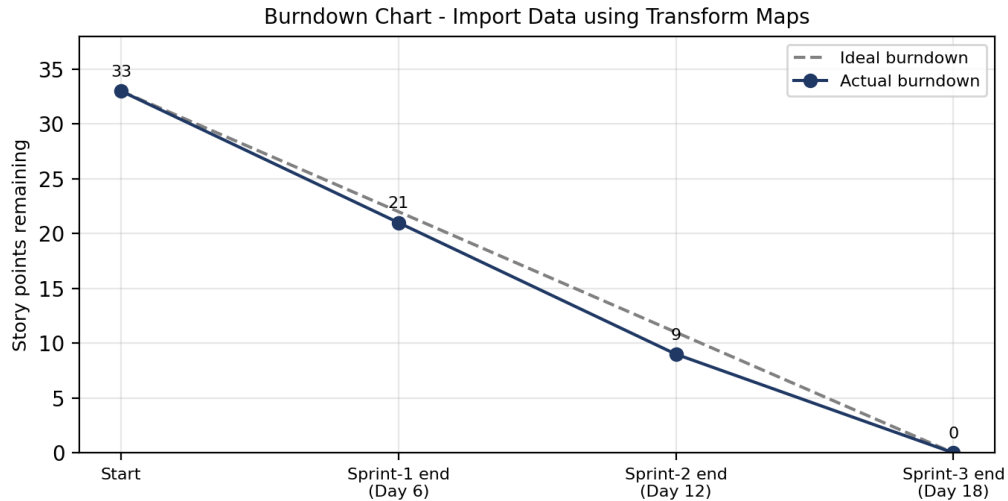


Figure 3: Burndown Chart

Checkpoint	Ideal Remaining	Actual Remaining	Variance
Start	33	33	0
Sprint-1 end (Day 6)	22	21	1 point ahead
Sprint-2 end (Day 12)	11	9	2 points ahead
Sprint-3 end (Day 18)	0	0	On target

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing focused on the functional behaviour of the import and the access rules, together with simple performance checks on the import. The test strategy covers positive scenarios (authorised access works) and negative scenarios (unauthorised access is blocked).

Parameter	Details
Testing type	Functional, security and import performance testing
Environment	ServiceNow instance, web browser
Test data	Spreadsheet with four columns: Training Name, Completion Date, Status, Employee
Test method	Manual execution with user impersonation
Testing period	28 Sep 2026 to 03 Oct 2026 (Sprint-3)

TC ID	Scenario	Test Steps	Expected Result	Result
TC-001	Table creation	Open the table and check the four fields	Training Name, Completion Date, Status and Employee are present	Pass
TC-002	Status choices	Open Status dropdown on a new record	Only In Progress and Completed appear	Pass
TC-003	Spreadsheet upload	Use Load Data and upload the file	Rows are placed in the import set table	Pass
TC-004	Field mapping	Open the Transform Map and review the mapping	Four source fields map to four target fields	Pass
TC-005	Run Transform	Click Transform on the import set	Records are inserted into Employee Training Records	Pass
TC-006	Employee link	Open an imported record	Employee shows as a reference to a user	Pass
TC-007	Department dot-walk	Add Employee.Department to the form	Department of the employee is displayed	Pass
TC-008	HR Manager access (positive)	Impersonate HR Manager and open the module	Records can be viewed and edited	Pass
TC-009	Unauthorised access (negative)	Impersonate a user without the role	Table and records are not visible	Pass
TC-010	Write restriction (negative)	Impersonate a user without the role and try to edit	Edit is denied	Pass

Test case analysis:

Section	Total Cases	Not Tested	Fail	Pass
Table and Status setup	2	0	0	2
Import and Transform	3	0	0	3

Section	Total Cases	Not Tested	Fail	Pass
Employee linking	2	0	0	2
Access control	3	0	0	3
Total	10	0	0	10

Defect analysis: No functional defects remained open at the time of release. Issues found during development were corrected within the same sprint.

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
Fixed	0	0	0	0	0
Open	0	0	0	0	0
Totals	0	0	0	0	0

Performance Check	Observation
Upload of spreadsheet	Completed in a single Load Data action
Transform execution	All staged rows processed in one run
Record count match	Rows in spreadsheet equal records in target table
Form load with dot-walked Department	Form opens normally without delay

7. RESULTS

7.1 Output Screenshots

This section presents the outputs of each stage of the project. Each screenshot is described by its caption, in the order the work was carried out.

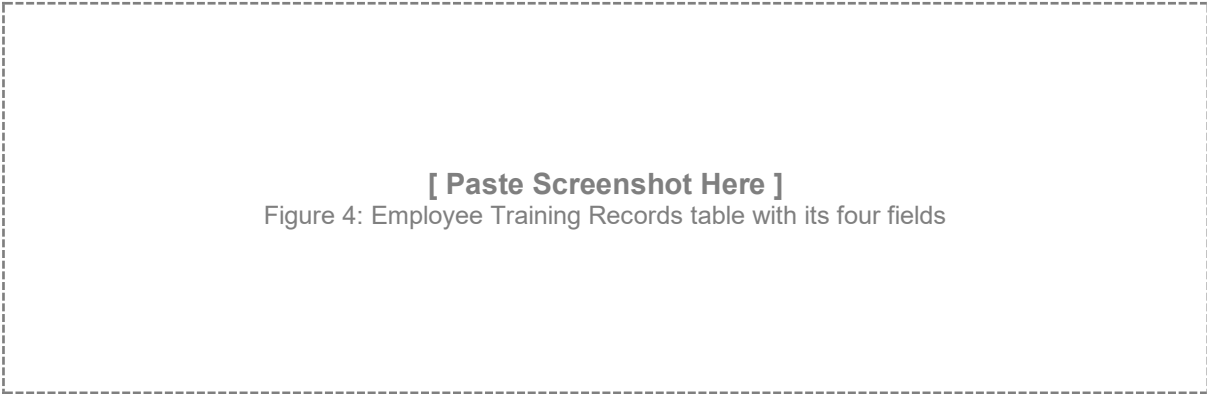


Figure 4: Employee Training Records table with its four fields

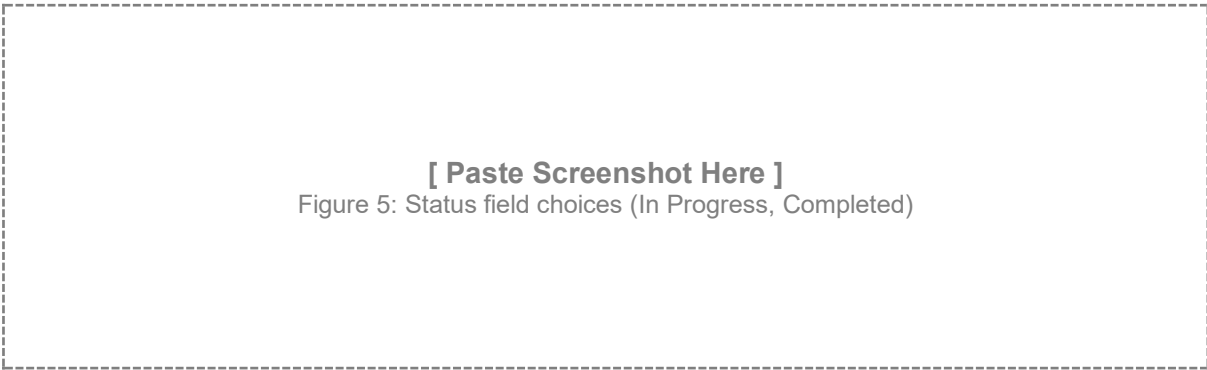


Figure 5: Status field choices (In Progress, Completed)

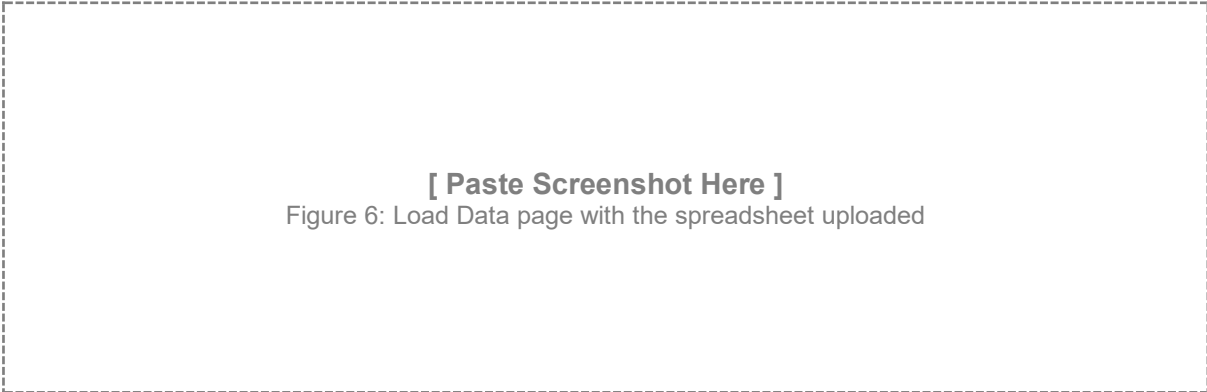


Figure 6: Load Data page with the spreadsheet uploaded

[Paste Screenshot Here]

Figure 7: Transform Map field mapping

Figure 7: Transform Map field mapping

[Paste Screenshot Here]

Figure 8: Imported records in the Employee Training Records list

Figure 8: Imported records in the Employee Training Records list

[Paste Screenshot Here]

Figure 9: Form showing Employee with dot-walked Department

Figure 9: Form showing Employee with dot-walked Department

[Paste Screenshot Here]

Figure 10: HR Manager role and backup user

Figure 10: HR Manager role and backup user

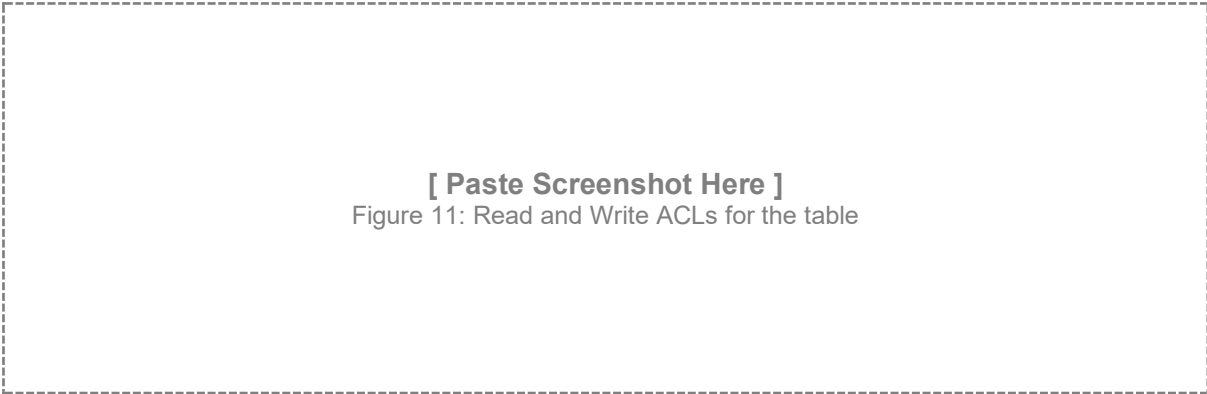


Figure 11: Read and Write ACLs for the table

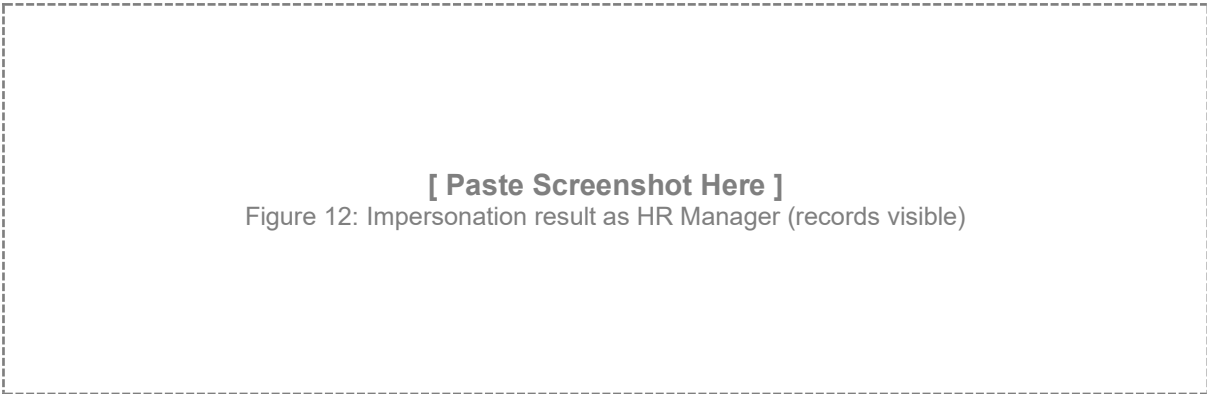


Figure 12: Impersonation result as HR Manager (records visible)

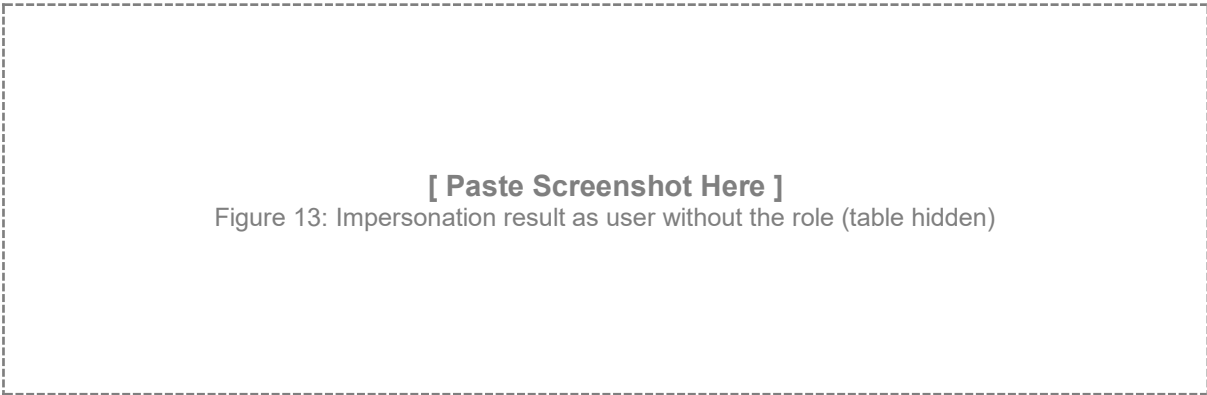


Figure 13: Impersonation result as user without the role (table hidden)

Expected Outcome	Result Achieved
Spreadsheet records appear in the target table	Achieved
Each record linked to an employee	Achieved
Department visible through dot-walk	Achieved
HR Manager can view and edit	Achieved
Other users cannot see the table	Achieved

8. ADVANTAGES & DISADVANTAGES

The table below compares the strengths and limitations of the implemented solution.

Advantages	Disadvantages
Bulk import saves significant time compared with manual entry	Spreadsheet columns must match the expected structure exactly
Transform Map reduces mapping and typing errors	Initial setup of the Transform Map needs administrator knowledge
Employee reference enables department-wise reporting	Employees must already exist in sys_user for the link to work
Role-based ACLs protect sensitive HR data	Elevating to security_admin is required to create ACLs, which needs careful control
No custom code is needed; uses standard ServiceNow features	Large or complex files may need additional validation rules
Mapping can be reused for later uploads	Duplicate records are not automatically prevented unless a coalesce field is set

Criteria	Manual Entry	Transform Map Import	Better Option
Speed	Slow	Fast	Transform Map
Accuracy	Error-prone	Consistent	Transform Map
Employee link	None	Reference field	Transform Map
Security	None	Role and ACLs	Transform Map
Setup effort	None	One-time setup	Manual Entry

9. CONCLUSION

The project successfully shows how spreadsheet data can be brought into ServiceNow in a fast, accurate and secure way. A custom Employee Training Records table was created, a spreadsheet was uploaded using Load Data, and a Transform Map loaded the data into the target table. Each record was linked to an employee and the Department was displayed using dot-walking.

Security was added through an HR Manager role and Read and Write ACLs, and impersonation testing confirmed that authorised users can view and edit the records while other users cannot see the table. All twelve user stories, amounting to 33 story points, were completed within the three planned sprints at an average velocity of 11 story points per sprint.

The solution meets the needs identified in the Problem-Solution Fit and replaces a slow, error-prone and unprotected manual process with a governed and repeatable one.

10. FUTURE SCOPE

The solution can be extended in several directions.

- **Scheduled imports:** Automate imports from a shared location so that new spreadsheets load without manual action.
- **Duplicate prevention:** Set coalesce fields in the Transform Map so repeated uploads update existing records.
- **Dashboards and reports:** Build department-wise training completion reports and dashboards.
- **Notifications:** Send reminders to employees whose training is still In Progress.
- **Validation scripts:** Add transform scripts to validate dates and reject unmatched employees.
- **More fields:** Add fields such as trainer, duration and certificate number.
- **Self-service:** Allow employees to view their own training history with a read-only role.

11. APPENDIX

Source code: Not applicable. The solution is built with standard ServiceNow configuration and requires no custom source code.

A. Spreadsheet Structure

Column	Target Field	Type	Notes
Training Name	Training Name	String	Name of the training
Completion Date	Completion Date	Date	Date the training was completed
Status	Status	Choice	In Progress or Completed
Employee	Employee	Reference (sys_user)	Linked to the employee record

B. Sample Data Format

Training Name	Completion Date	Status	Employee
Data Security Basics	2026-08-12	Completed	Employee 1
Workplace Safety	2026-09-03	Completed	Employee 2
Leadership Essentials		In Progress	Employee 3

Sample rows shown only to illustrate the format

C. Access Control Summary

Operation	Role Required	Applies To
Read	HR Manager	Employee Training Records table
Write	HR Manager	Employee Training Records table
ACL creation	security_admin (elevated)	Administrator

D. Implementation Steps Summary

1. Create the Employee Training Records table with four fields.
2. Add In Progress and Completed choices to Status.
3. Prepare the spreadsheet with matching columns.
4. Upload the file through Load Data into an import set table.
5. Create a Transform Map and map the four fields.
6. Run Transform to load the target table.
7. Dot-walk Employee to Department on the form.
8. Create the HR Manager role and a backup user and assign the role.
9. Add the role to the application menu and module.
10. Elevate to security_admin and create Read and Write ACLs.
11. Impersonate users to verify access.

E. Glossary

Term	Meaning
Load Data	ServiceNow feature to upload a file into an import set table
Import Set	Staging table that temporarily holds uploaded rows
Transform Map	Rules that map source fields to target table fields
Dot-walking	Reaching a related record's field through a reference field, such as Employee.Department
ACL	Access Control List rule that allows or denies access to data
Impersonation	Temporarily acting as another user to test their access